

HCF and LCM

LCM (Least common multiple) of two or more given numbers is the least number that is exactly divisible by each of them.

HCF (Highest common factor) of two or more numbers is the greatest number that divides each of them exactly. HCF is also known as the 'Highest common Divisor' (HCD) and the Greatest Common Measure (GCM).

The concept of multiples and factors

→ If X, Y, and Z are three natural numbers and $X \times Y = Z$, then

- X and Y are called the factors of Z.
- Z is said to be divisible by X and Y.
- Z is said to be a multiple of X and Y.

Example: The set of positive Integers which are factors of 18 is (1, 2, 3, 6, 9, 18).

Basic Concepts of H.C.F. and LCM

Method of finding H.C.F.

To find the HCF of the given numbers

- Break the given numbers into their prime factors.
- The HCF will be the product of all the prime factors common to all the numbers.

Let us learn the process of finding HCF with the help of some solved examples.

Example:-

Find the HCF of 96, 36 and 18 ?

Solution:-

$$96 = 2 \times 3 \times 2 \times 2 \times 2 \times 2$$

$$36 = 2 \times 3 \times 2 \times 3$$

$$18 = 2 \times 3 \times 3$$

Therefore, the HCF of 96, 36 and 18 is the product of the highest number of common factors in the given numbers i.e., $2 \times 3 = 6$. In other words, 6 is the largest possible integer, which can divide 96, 36 and 18 without leaving any remainder.

Example:- Find the H.C.F. of 42 and 70 ?

Solution:-

$$42 = 3 \times 2 \times 7$$

$$70 = 5 \times 2 \times 7$$

Hence, H.C.F. of 42 and 70 = $2 \times 7 = 14$.

HCF by Division Method

Example:- HCF of 24, 48, 72, and 100.

Solution:- To start the division method select the smallest two numbers./

$$\begin{array}{r} 24 \overline{) 48} \begin{array}{l} 2 \\ -48 \\ \hline \end{array} \quad \begin{array}{r} 48 \overline{) 72} \begin{array}{l} 1 \\ -48 \\ \hline \end{array} \quad \begin{array}{r} 24 \overline{) 100} \begin{array}{l} 4 \\ -96 \\ \hline \end{array} \end{array}$$

$$\text{HCF of 24 and 48} = 24$$

$$\text{HCF of 24, 48 and 72} = 24$$

$$\text{HCF of 24, 48, 72, and 100} = 4.$$

Example:- HCF of 1785, 1995, 3381.

Solution:-

$$\begin{array}{r} 1785 \overline{) 1995} \begin{array}{l} 1 \\ -1785 \\ \hline \end{array} \quad \begin{array}{r} 105 \overline{) 3381} \begin{array}{l} 32 \\ -315 \\ \hline \end{array} \end{array}$$

$$\text{HCF of 1785 and 1995} = 105$$

$$\text{HCF of 1785, 1995 and 3381} = 21$$

NOTE :-

- HCF of two prime numbers is always 1.
- HCF of co-prime numbers is always 1.

Method of finding L.C.M

The Least Common Multiple of two or more numbers is the smallest number which is exactly divisible by all of them. In other words, it is the product of the highest powers of all the prime factors of the given numbers.

To find the LCM of given numbers:

- Break the given numbers into their prime factors.
- The LCM will be the product of the highest power of all the factors that occur in the given numbers.

Let us take some solved examples.

Example:- Find the LCM of 96, 36 and 18.

$$\text{Solution: } 96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3$$

$$= 2^5 \times 3^1; 36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$$

$$18 = 2 \times 3 \times 3 = 2^1 \times 3^2$$

Therefore, LCM of 96, 36 and 18 is the product of the highest powers of all the prime factors, i.e. $2^5 \times 3^2 = 32 \times 9 = 288$. That is, 288 is the smallest integer which is divisible by 96, 36 and 18 without leaving any remainder.

Example:- Find the LCM of 42 and 70

Solution:-

$$42 = 3 \times 2 \times 7$$

$$70 = 5 \times 2 \times 7$$

$$\text{Hence, LCM is } 2 \times 3 \times 5 \times 7 = 210.$$

Example:- LCM of 6, 12, 8 ?

Solution:-

$$\begin{array}{c|c} 2 & 6, 12, 8 \\ 2 & 3, 6, 4 \\ 3 & 3, 3, 2 \\ 2 & 1, 1, 2 \\ & 1, 1, 1 \end{array}$$

$$\text{LCM} = 2 \times 2 \times 3 \times 2 = 24$$

HCF of 6, 12, 18

Firstly find out the factors of 6, 12, 18 and then multiply the common factors.

$$\begin{array}{c|c} 2 & 6 \\ 3 & 3 \\ \hline & 1 \end{array} \quad \begin{array}{c|c} 2 & 12 \\ 2 & 6 \\ 3 & 3 \\ \hline & 1 \end{array} \quad \begin{array}{c|c} 2 & 18 \\ 3 & 9 \\ 3 & 3 \\ \hline & 1 \end{array}$$

$$6 = 2 \times 3, 12 = 2 \times 2 \times 3, 18 = 2 \times 3 \times 3$$

$$\text{HCF} = 2 \times 3 = 6$$

Try finding HCF and LCM of 3, 6, 9, 12 yourself. HCF of 3, 6, 9, 12 can also be found by Division method. It is useful when the numbers are bigger.

NOTE :-

- HCF of A, B and C is the highest divisor which can exactly divide A, B, and C.
- LCM of A, B and C is the lowest dividend which is exactly divisible by A, B, and C.

There is one very important relationship, given below, between two numbers and their HCF and LCM. Many problems have appeared in various competitive exams based on this relationship.

Important Concepts:-

(1).

$$\text{LCM} \times \text{HCF} = 1^{\text{st}} \text{ number} \times 2^{\text{nd}} \text{ number}$$

Example:- For numbers 8 and 12,

$$\text{LCM} = 24 \text{ and } \text{HCF} = 4$$

$$\text{Now, } \text{LCM} \times \text{HCF} = 24 \times 4 = 96$$

$$\text{also, } 8 \times 12 = 96$$

(2). HCF of some numbers is always a factor of LCM of the numbers.

(3).

$$\text{LCM of Fraction} = \frac{\text{LCM of numerator}}{\text{HCF of denominator}}$$

(4).

$$\text{HCF of Fraction} = \frac{\text{HCF of numerator}}{\text{LCM of denominator}}$$

Example: LCM and HCF of $\frac{1}{2}, \frac{2}{3}$ and $\frac{3}{4}$

$$\begin{aligned} \text{Solution:- } \text{LCM} &= \frac{\text{LCM of numerator}}{\text{HCF of denominator}} \\ &= \frac{\text{LCM of } 1, 2, 3}{\text{HCF of } 2, 3, 4} = \frac{6}{1} \end{aligned}$$

$$\begin{aligned} \text{HCF} &= \frac{\text{HCF of numerator}}{\text{LCM of denominator}} \\ &= \frac{\text{HCF of } 1, 2, 3}{\text{LCM of } 2, 3, 4} = \frac{1}{12} \end{aligned}$$

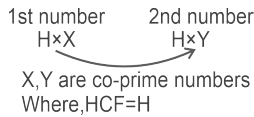
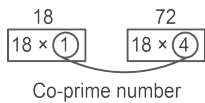
(5) Co- Prime numbers :

If the HCF of two numbers is 1 then they are called co-prime numbers.

$$(6) \quad \frac{\text{LCM}}{\text{HCF}} = \text{Product};$$

where LCM and HCF are of two numbers N_1 and N_2 . If we find two co-prime factors, F_1 and F_2 , of the Product as obtained above.

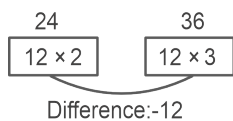
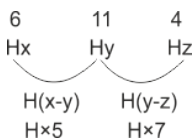
(7)

**Example:-**

(8).

If 1st number = N_1 and 2nd number = N_2 HCF of $N_1 = Hx$ HCF of $N_2 = Hy$ And HCF of N_1 and $N_2 = H$

So,

Difference between N_1 and $N_2 = Hx - Hy$ $= H(x - y)$ **Example :-** $N_1 = 24$ and $N_2 = 36$ **Note :-** HCF is always either the difference of two numbers or factors of difference of two numbers.

(9). When the second divisor is a factor of the first divisor, then the second remainder is obtained by dividing the first remainder by the second divisor.

Example :- When 29 is divided by 8, the remainder obtained is 5, then what will be the remainder when the same number is divided by 4?**Solution:-** Here, the second divisor that is 4 is a factor of the first divisor that is 8. So, on dividing the first remainder that is 5 by the second divisor that is 4, we get our second remainder which is 1. So, the required answer is 1.

Variety Questions

Q.1. The LCM of $x^2 - 8x + 15$ and $x^2 - 5x + 6$ is:

SSC CPO 05/10/2023 (2nd Shift)

- (a) $(x + 5)(x + 2)(x + 3)$
 (b) $(x - 5)(x - 2)(x - 3)$
 (c) $(x + 5)(x - 2)(x - 3)$
 (d) $(x - 2)(x - 3)^2(x - 5)$

Q.2. The greatest possible length that can be used to measure exactly the lengths of 3 m 15 cm, 5 m, and 6 m 85 cm is:

SSC CPO 05/10/2023 (2nd Shift)

- (a) 11 cm (b) 7 cm (c) 9 cm (d) 5 cm

Q.3. What is the HCF of $(x^6 + 1)$ and $(x^4 - 1)$ SSC CPO 05/10/2023 (1st Shift)

- (a) $(1 + x^2)$ (b) $(1 + x)$ (c) 1 (d) $(1 - x^2)$

Q.4. The LCM of two prime numbers x and y ($x > y$) is 533. The value of $4y - x$ is:

SSC CPO 03/10/2023 (3rd Shift)

- (a) 11 (b) 21 (c) 18 (d) 23

Q.5. What is the HCF of the polynomials $(x^3 - 8)$, $(x^3 - 6x^2 + 12x - 8)$ and $(x^3 - 4x^2 + 4x)$?

SSC MTS 12/09/2023 (3rd Shift)

- (a) $(x - 1)$ (b) $(x - 2)$ (c) $(x - 8)$ (d) $(x - 4)$

Q.6. Radha, Pratima and Reena begin to jog around a circular path and they complete their revolutions in 50 seconds, 75 seconds and 100 seconds, respectively. After how much time (in minutes) will they meet together at the starting point for the first time?

SSC MTS 11/09/2023 (3rd Shift)

- (a) 6 (b) 5 (c) 4 (d) 3

Q.7. Sanjay wants to exactly measure the lengths 7000 mm, 3850 mm and 12950 mm using a single measuring tape. What will be the greatest possible length (in cm) of the measuring tape?

SSC MTS 11/09/2023 (2nd Shift)

- (a) 40 (b) 45 (c) 35 (d) 30

Q.8. Two tankers contain 850 litres and 680 litres of oil. Find the maximum capacity of a container which can measure the oil of both tankers when used, an exact number of times.

SSC MTS 06/09/2023 (2nd Shift)

- (a) 425 litres (b) 680 litres
 (c) 340 litres (d) 170 litres

Q.9. Find the greatest two digit number which on dividing 219, 365, 511 leaves the remainders 3, 5, 7, respectively.

SSC MTS 01/09/2023 (3rd Shift)

- (a) 63 (b) 82 (c) 53 (d) 72

Q.10. Five bells commence tolling together and toll at intervals of 5, 10, 15, 20 and 25 seconds respectively. In 1 hour 20 minutes, how many times do they toll together?

SSC MTS 17/05/2023 (Afternoon)

- (a) 20 (b) 15 (c) 14 (d) 17

Q.11. There is a circular path around a sports field. Rahul takes 15 minutes to drive one round of the field, while Anil takes 18 minutes for the same. Suppose they both start from the same point and at the same time, and go in the same direction. After how many minutes will they meet again at the starting point?

SSC MTS 11/05/2023 (Afternoon)

- (a) 120 (b) 100 (c) 80 (d) 90

Q.12. Let $x = 224$ and, $y = 322$. If the highest common factor of $23x$ and $a \times y$ is divisible by x and y , then what can be the possible value of a ?

SSC CPO 11/11/2022 (Evening)

- (a) 16 (b) 8 (c) 12 (d) 4

Q.13. The highest common factor of 108, 72 and $5a$ is a . What can be the least common multiple of 108, 72 and a ?

SSC CPO 10/11/2022 (Afternoon)

- (a) 432 (b) 324 (c) 108 (d) 216

Q.14. What is the greatest positive integer that divides 554, 714 and 213 leaving the remainder 43, 57 and 67, respectively?

SSC CPO 10/11/2022 (Afternoon)

- (a) 95 (b) 71 (c) 83 (d) 73

Q.15. Find the HCF of $(4^{315} - 1)$ and $(4^{25} - 1)$.

SSC MTS 26/07/2022 (Morning)

- (a) 1 (b) $(4^{25} - 1)$ (c) 1024 (d) 1023

Q.16. 120 apples, 240 oranges and 150 pears are packed in cartons in such a way that each carton has the same number of fruits, each carton contains only one type of fruit and fruit is left unpacked. What is the smallest possible number of cartons needed for the purpose?

SSC MTS 08/07/2022 (Evening)

- (a) 50 (b) 40 (c) 17 (d) 30

Q.17. The traffic lights at 3 different road crossings change after every 48 sec, 72 sec and 108 sec, respectively. If they all change simultaneously at 8 : 20 a.m., then at what will they next change again simultaneously?

SSC MTS 07/07/2022 (Morning)

- (a) 8 : 27 : 12 a.m. (b) 8 : 33 : 32 a.m.
 (c) 8 : 12 : 18 a.m. (d) 8 : 40 : 14 a.m.

Q.18. A and B are two prime numbers such that $A > B$ and their LCM is 209. The value of $A^2 - B$ is?

SSC CGL 12/04/2022 (Evening)

- (a) 350 (b) 372 (c) 361 (d) 339

Q.19. 13, a, b, c are four distinct numbers and the HCF of each pair of numbers (13, a) : (13, b) : (13, c) is 13, where a, b, c are each less than 60 and $a < b < c$. What is the value of $\frac{a+c}{b}$?

SSC CGL 13/04/2021 (Morning)

- (a) 3.5 (b) 2 (c) 5 (d) 4.5

Q.20. The sum of two numbers is 1215 and their HCF is 81. If the numbers lie between 500 and 700, then the sum of reciprocals of the numbers is

SSC CPO 13/12/2019 (Evening)

(a) $\frac{5}{1512}$ (b) $\frac{5}{378}$ (c) $\frac{5}{702}$ (d) $\frac{5}{1188}$

Q.21. If r is the remainder when each of 6454, 7306 and 8797 is divided by the greatest number d ($d > 1$), then $(d - r)$ is:
SSC CPO 13/12/2019 (Morning)
(a) 126 (b) 64 (c) 137 (d) 149

Q.22. In finding the HCF of two numbers by division method, the quotients are 1, 8 and 2 respectively, and the last divisor is 105, what is the sum of the numbers?
SSC CPO 11/12/2019 (Evening)
(a) 3570 (b) 3885 (c) 3780 (d) 3675

Q.23. When the smallest number x is divided by 5, 6, 8, 9 and 12, it gives remainder 1 in each case. But x is divisible by 13. What will be the remainder when x will be divided by 31
SSC MTS 20/08/2019 (Afternoon)
(a) 1 (b) 5 (c) 3 (d) 0

Q.24. The Highest Common Factor and Lowest Common Multiple of two numbers p and q are A and B respectively. If $A + B = p + q$, then the value of $A^3 + B^3$ is:
SSC MTS 09/08/2019 (Evening)
(a) p^3 (b) q^3 (c) $p^3 + q^3$ (d) $p^3 - q^3$

Q.25. What is the largest number that divides 460, 491 and 553 and leaves the remainder 26 in each case?
SSC MTS 06/08/2019 (Afternoon)
(a) 27 (b) 35 (c) 33 (d) 31

Q.26. A is the smallest three-digit number which when divided by 3, 4 and 5 gives remainder 1, 2 and 3 respectively. What is the sum of the digits of A ?
SSC MTS 05/08/2019 (Afternoon)
(a) 11 (b) 10 (c) 6 (d) 8

Q.27. The product of two numbers is 6760 and their HCF is 13. How many such pairs of numbers can be formed?
SSC CPO 16/03/2019 (Evening)
(a) 2 (b) 3 (c) 1 (d) 4

Q.28. An oil merchant has 3 varieties of oil of volumes 432, 594 and 702 respectively. The number of cans of equal size that would be required to fill the oil separately is:
SSC CPO 16/03/2019 (Afternoon)
(a) 13, 15, 17 (b) 8, 11, 13
(c) 8, 13, 15 (d) 6, 9, 11

Q.29. Which of the following statements is true?
SSC CPO 14/03/2019 (Morning)
(a) LCM of two natural numbers is divisible by their HCF.
(b) HCF + LCM of two numbers =

Product of the two numbers.

- (c) Two prime numbers are co-prime numbers if their LCM is 1.
(d) HCF of two numbers is the smallest common divisor of both numbers.

Practice Questions

SSC CHSL 2024 Tier - 2

Q.30. A milk vendor has 75 litres of cow milk, 45 litres of toned milk and 30 litres of full cream milk. These should be placed in different trays such that each tray contains the same quantity of milk and two kinds of milk should not be mixed in a tray. What is the least number of trays required to place all the milk?
SSC CHSL Tier II (18/11/2024)
(a) 15 (b) 25 (c) 10 (d) 20

Q.31. Three numbers are in the ratio 5 : 7 : 9, and their LCM is 1260. What is the difference between the largest and the smallest number?
SSC CHSL Tier II (18/11/2024)
(a) 24 (b) 20 (c) 16 (d) 32

SSC MTS 2024 Tier - 1

Q.32. Three red lights on the same road start blinking at the same time and blink repeatedly after every 4, 6 and 8 minutes, respectively. How many times will they blink together in an interval of 3 hours, excluding blinking together at the starting moment?
SSC MTS 08/10/2024 (Afternoon)
(a) 5 (b) 7 (c) 6 (d) 4

Q.33. As part of the partial work from home process, Amita went to office every 2nd day, Amrita went to office every 3rd day, and Tosita went to office every 6th day, respectively. All three went to office on 31 October 2021. On how many days did all three attend office on the same date during the remaining two months of the year?
SSC MTS 15/10/2024 (Evening)
(a) 9 (b) 10 (c) 11 (d) 12

Q.34. Find the least number which when divided by 20, 25, 35 and 40 leaves remainders 14, 19, 29 and 34, respectively.
SSC MTS 16/10/2024 (Evening)
(a) 1284 (b) 1294 (c) 1384 (d) 1394

Q.35. If two positive integers x and y are written as $x = a^2b^3$ and $y = a^3b$; and a and b are prime numbers, then HCF (x, y) is:
SSC MTS 17/10/2024 (Evening)
(a) ab^2 (b) a^2b^2 (c) ab (d) a^2b

Q.36. x, y and z are divisible by 2, 3 and 4, respectively. What is the highest common factor (HCF) of $6x, 4y$ and $3z$?
SSC MTS 18/10/2024 (Morning)
(a) At least 36 (b) At least 12
(c) At least 18 (d) At least 16

Q.37. Find the HCF of:
 $3^3 \times 11^2, 2^4 \times 3^2 \times 5^3 \times 11^3$ and $3 \times 5 \times 7 \times 11$
SSC MTS 18/10/2024 (Evening)
(a) 385 (b) 1815 (c) 33 (d) 165

Q.38. Two numbers are in the ratio of 7 : 9 and their LCM is 630. Find the numbers
SSC MTS 21/10/2024 (Morning)
(a) 56, 72 (b) 77, 99 (c) 70, 90 (d) 63, 81

Q.39. If the HCF of 156 and 96 is expressible in the form of $156 \times 12 + 96A$, then the value of A is:
SSC MTS 23/10/2024 (Afternoon)
(a) -19.735 (b) 19.735
(c) -19.375 (d) 19.375

Q.40. The HCF of 1.75, 5.6 and 7 is:
SSC MTS 28/10/2024 (Morning)
(a) 0.35 (b) 3.5 (c) 0.07 (d) 0.7

Q.41. The LCM of $x^3 - 27, x^2 - x - 6, x + 2$ is:
SSC MTS 04/11/2024 (Morning)
(a) $(x - 3)(x + 2)(x^2 - 3x + 9)$
(b) $(x + 3)(x + 2)(x^2 - 3x + 9)$
(c) $(x - 3)(x + 2)(x^2 + 3x - 9)$
(d) $(x - 3)(x + 2)(x^2 + 3x + 9)$

Q.42. The product of LCM and HCF of two numbers is 24. If the difference between the two numbers is 2, then find the numbers.
SSC MTS 13/11/2024 (Evening)
(a) 5 and 3 (b) 6 and 4
(c) 2 and 4 (d) 8 and 6

Q.43. The LCM of $\frac{1}{6}, \frac{4}{15}, \frac{5}{9}, \frac{7}{27}, \frac{8}{3}$ is:
SSC MTS 14/11/2024 (Afternoon)
(a) $92\frac{1}{3}$ (b) $93\frac{2}{3}$ (c) $93\frac{1}{3}$ (d) $92\frac{2}{3}$

SSC CPO 2023 Tier - 1

Q.44. The largest number of four digits that is exactly divisible by 17 and 36 is:
SSC CPO 03/10/2023 (1st Shift)
(a) 8568 (b) 9180 (c) 9792 (d) 7956

Q.45. Two numbers are in the ratio 3 : 4. The product of their HCF and LCM is 2700. The sum of the numbers is:
SSC CPO 03/10/2023 (1st Shift)
(a) 45 (b) 105 (c) 60 (d) 15

Q.46. Find the greatest number which

when divides 261, 853 and 1221, leaves a remainder of 5 in each case.

SSC CPO 03/10/2023 (2nd Shift)

(a) 18 (b) 17 (c) 16 (d) 19

Q.47. Find the LCM of $\frac{3}{2}$, $\frac{81}{16}$ and $\frac{9}{8}$.

SSC CPO 03/10/2023 (2nd Shift)

(a) $\frac{111}{2}$ (b) $\frac{91}{2}$ (c) $\frac{81}{2}$ (d) $\frac{101}{2}$

Q.48. Find the greatest possible length (in metres) that can be used to exactly measure the lengths 6 m, 5 m 25 cm and 12 m 50 cm.

SSC CPO 03/10/2023 (2nd Shift)

(a) 0.35 m (b) 0.90 m (c) 0.75 m (d) 0.25 m

Q.49. The least number which should be added to 1351 so that the sum is exactly divisible by 2, 4, 6 and 8 is:

SSC CPO 03/10/2023 (3rd Shift)

(a) 13 (b) 11 (c) 15 (d) 17

Q.50. Let the HCF of m and n be 'a' and let $n = ab$, LCM of m and n is given by:

SSC CPO 04/10/2023 (1st Shift)

(a) ab (b) am (c) bm (d) mn

Q.51. The LCM and HCF of two numbers are 1105 and 5. If the LCM is 17 times the first number, then find the two numbers.

SSC CPO 04/10/2023 (2nd Shift)

(a) 55 and 85 (b) 65 and 75

(c) 60 and 80 (d) 65 and 85

Q.52. If the HCF of 45 and 55 is expressible in the form of $55 \times 5 + 45m$, then what is the value of m?

SSC CPO 04/10/2023 (3rd Shift)

(a) 5 (b) -6 (c) -5 (d) 6

Q.53. Find the least number which when divided by 4, 9, 12 and 15, leaves the remainder 3 in each case.

SSC CPO 04/10/2023 (3rd Shift)

(a) 193 (b) 183 (c) 360 (d) 180

Q.54. Determine the LCM of two numbers if their HCF is 12 and their ratio is 13 : 15.

SSC CPO 04/10/2023 (3rd Shift)

(a) 2450 (b) 1780 (c) 1890 (d) 2340

Q.55. Which of the following numbers leaves the remainder equal to the highest common factor of 6, 8 and 9, when divided by 6, 8 and 9?

SSC CPO 04/10/2023 (3rd Shift)

(a) 506 (b) 575 (c) 291 (d) 433

Q.56. Find the least number which is exactly divisible by 20, 28, 34, 60 and 75.

SSC CPO 05/10/2023 (1st Shift)

(a) 34500 (b) 35900 (c) 35700 (d) 36220

Q.57. Which is the largest number that

divides each of 1036, 1813 and 3885 without leaving any remainder?

SSC CPO 05/10/2023 (1st Shift)

(a) 259 (b) 111 (c) 333 (d) 37

Q.58. What is the LCM of 0.15, 0.18 and 0.45?

SSC CPO 05/10/2023 (2nd Shift)

(a) 0.6 (b) 0.9 (c) 0.81 (d) 0.09

Q.59. What will be the least number which when doubled will be exactly divisible by 12, 14, 16 and 18?

SSC CPO 05/10/2023 (2nd Shift)

(a) 636 (b) 226 (c) 428 (d) 504

Q.60. The LCM of two numbers is five times their HCF. If the product of the two numbers is 20480, then find their HCF and LCM, respectively.

SSC CPO 05/10/2023 (3rd Shift)

(a) 64 and 320 (b) 56 and 280

(c) 48 and 240 (d) 46 and 230

Q.61. If the sum of two numbers is 60 and their HCF and LCM are 5 and 60, respectively, then the sum of the reciprocals of the numbers will be:

SSC CPO 05/10/2023 (3rd Shift)

(a) $\frac{1}{4}$ (b) $\frac{1}{5}$ (c) $\frac{1}{11}$ (d) $\frac{1}{6}$

Q.62. The least number of five digits which is exactly divisible by 9, 12, 15, 25 and 27 is:

SSC CPO 05/10/2023 (3rd Shift)

(a) 10250 (b) 10800 (c) 10600 (d) 10700

SSC MTS 2023 Tier - 1

Q.63. What is the largest number that divides 627, 15630 and 3128 and leaves remainders of 2, 5 and 3, respectively?

SSC MTS 04/09/2023 (3rd Shift)

(a) 775 (b) 650 (c) 625 (d) 1225

Q.64. The HCF of three numbers is 57. If they are in the ratio of 4 : 5 : 6, then find the numbers.

SSC MTS 05/09/2023 (2nd Shift)

(a) 228, 285, 342 (b) 236, 295, 354

(c) 240, 300, 360 (d) 232, 290, 348

Q.65. Three numbers are in the ratio 3 : 5 : 7. If their LCM is 2625, find their HCF.

SSC MTS 05/09/2023 (3rd Shift)

(a) 100 (b) 125 (c) 50 (d) 25

Q.66. The largest three-digit number which gives the same remainder 2 when divided by 3, 5 and 9 is _____.

SSC MTS 08/09/2023 (1st Shift)

(a) 980 (b) 992 (c) 995 (d) 990

Q.67. The HCF of two numbers is 12. Which of the following can never be their LCM?

SSC MTS 12/09/2023 (1st Shift)

(a) 24 (b) 74 (c) 48 (d) 60

Q.68. The product of two numbers is 4107. If the HCF of these numbers is 37, then the greater number is:

SSC MTS 12/09/2023 (2nd Shift)

(a) 111 (b) 74 (c) 185 (d) 37

Q.69. The LCM of the two numbers is 1440 and their HCF is 32. If the first number is 288, then find the second number.

SSC MTS 13/09/2023 (2nd Shift)

(a) 150 (b) 145 (c) 180 (d) 160

Q.70. What is the greatest three-digit number which when divided by 4, 8 and 9, leaves no remainder?

SSC MTS 13/09/2023 (3rd Shift)

(a) 924 (b) 963 (c) 932 (d) 936

Q.71. Find the smallest three digit number which when added to 5 is exactly divisible by 12, 14 and 18.

SSC MTS 14/09/2023 (2nd Shift)

(a) 240 (b) 225 (c) 245 (d) 247

Q.72. The ratio of two numbers is 3 : 4. If the sum of these two numbers is 70, then what is their LCM?

SSC MTS 14/09/2023 (3rd Shift)

(a) 120 (b) 96 (c) 108 (d) 84

SSC CGL 2023 Tier - 1

Q.73. Find the least number divisible by 2, 3, 5, 6, 9 and 18, which is a perfect square.

SSC CGL 24/07/2023 (2nd shift)

(a) 900 (b) 400 (c) 144 (d) 3600

SSC Selection Post (Phase - XI)

Q.74. The smallest six-digit number which is divisible by 12, 16, 24, 28 is:

Matriculation Level 30/06/2023 (Shift - 3)

(a) 100128 (b) 100190

(c) 100180 (d) 100160

SSC CHSL 2022 Tier - 2

Q.75. The HCF of two numbers is one-twentieth of their LCM. If one of the numbers is 96 and the difference of the LCM and the HCF is 456, then what is the other number?

SSC CHSL Tier II (26/06/2023)

(a) 48 (b) 120 (c) 144 (d) 72

SSC MTS 2022 Tier - 1

Q.76. What is the HCF of two prime numbers X and Y?

SSC MTS 02/05/2023 (Morning)

(a) 1 (b) 2 (c) Y (d) X

Q.77. What is that least number that should be added to 478 so that the number obtained becomes exactly divisible by 5, 6 and 12 ?

SSC MTS 04/05/2023 (Morning)

- (a) 62 (b) 12 (c) 2 (d) 52

Q.78. The product of two co-prime numbers is 483. Find the Least Common Multiple of both numbers.

SSC MTS 08/05/2023 (Afternoon)

- (a) 483 (b) 21 (c) 1 (d) 23

Q.79. What is that least perfect square which is exactly divisible by each of 14, 84 and 28 ?

SSC MTS 10/05/2023 (Afternoon)

- (a) 7056 (b) 3528 (c) 1764 (d) 441

Q.80. The greatest number which when divides 456 and 553 leaves the remainder as 6 and 3 respectively is:

SSC MTS 12/05/2023 (Evening)

- (a) 5 (b) 50 (c) 10 (d) 30

Q.81. The least number which when divided by 14, 18 and 36 and leaves 1 as remainder in each case is:

SSC MTS 16/05/2023 (Morning)

- (a) 250 (b) 252 (c) 253 (d) 251

Q.82. What will be the ratio of the Least Common Multiple and Highest Common Factor of 12, 24 and 36?

SSC MTS 13/06/2023 (Afternoon)

- (a) 3 : 1 (b) 4 : 1 (c) 6 : 1 (d) 2 : 1

Q.83. What is the least number of students in a class such that they can make rows of 10, 15 and 20 and also the total number of students is a perfect square?

SSC MTS 15/06/2023 (Morning)

- (a) 6400 (b) 1600 (c) 900 (d) 1800

Q.84. What is the HCF of 513 and 1017 ?

SSC MTS 16/06/2023 (Evening)

- (a) 15 (b) 8 (c) 9 (d) 13

Q.85. The Least Common Multiple of three different numbers is 40. Which of the following cannot be their Highest Common Factor ?

SSC MTS 19/06/2023 (Morning)

- (a) 20 (b) 25 (c) 5 (d) 10

Q.86. What is the smallest perfect square which is divisible by both 8 and 12 ?

SSC MTS 20/06/2023 (Afternoon)

- (a) 100 (b) 144 (c) 121 (d) 196

SSC CGL 2022 Tier - 2

Q.87. The product of the two numbers is 1500 and their HCF is 10. The number of such possible pairs is/are:

SSC CGL Tier II (02/03/2023)

- (a) 1 (b) 3 (c) 4 (d) 2

Q.88. If the highest common factor (HCF) of x and y is 15, then the HCF of $36x^2 - 81y^2$ and $81x^2 - 9y^2$ is divisible by

SSC CGL Tier II (06/03/2023)

- (a) 135 (b) 120 (c) 180 (d) 90

SSC CGL 2022 Tier - 1

Q.89. Calculate the HCF of $\frac{12}{5}$, $\frac{14}{15}$ & $\frac{16}{17}$

SSC CGL 03/12/2022 (1st Shift)

- (a) $\frac{4}{255}$ (b) $\frac{3}{255}$ (c) $\frac{2}{255}$ (d) $\frac{1}{255}$

Q.90. Choose the correct statement from the following.

SSC CGL 06/12/2022 (2nd Shift)

- (a) HCF is the least common multiple of the given numbers.
(b) HCF of two or more numbers is the highest number which perfectly divides all the given numbers.
(c) HCF is also called the least common divisor.
(d) In prime factorisation method of HCF, the multiples of all the given numbers are listed.

SSC CPO 2022 Tier - 1

Q.91. The least common multiple of a and b is 42. The LCM of $5a$ and $11b$ is :

SSC CPO 09/11/2022 (Morning)

- (a) 2310 (b) 4620 (c) 210 (d) 462

Q.92. The sum of two numbers is 1224 and their HCF is 68. The number of pairs of numbers satisfying the above condition is:

SSC CPO 09/11/2022 (Afternoon)

- (a) 3 (b) 4 (c) 6 (d) 2

Q.93. The LCM and ratio of three numbers are 1386 and 3 : 7 : 11, respectively. The sum of the greatest and least numbers is:

SSC CPO 11/11/2022 (Evening)

- (a) 60 (b) 64 (c) 84 (d) 108

SSC MTS 2021 Tier - 1

Q.94. If $P = 2^8 \times 3^5$, $Q = 2^3 \times 3^4$, and $R = 3^5 \times 2^7$, then what is the highest common factor of P , Q and R ?

SSC MTS 06/07/2022 (Afternoon)

- (a) $2^3 \times 3^4$ (b) $2^8 \times 3^5$
(c) $2^2 \times 3^2$ (d) $2^4 \times 3^5$

Q.95. The ratio of the LCM of two numbers to the sum of the same two numbers is 12 : 7. If their HCF is 4, what

is the product of these two numbers?

SSC MTS 14/07/2022 (Morning)

- (a) 192 (b) 172 (c) 196 (d) 169

Q.96. If K_1 and K_2 are two distinct prime numbers, then what is the product of the highest common factor and the least common multiple of K_1 and K_2 ?

SSC MTS 20/07/2022 (Afternoon)

- (a) 1 (b) $\frac{K_1}{K_2}$ (c) $K_1 + K_2$ (d) $K_1 \times K_2$

Q.97. Find the smallest natural number 'x' that must be subtracted from 1800 so that $(1800 - x)$, when divided by 7, 11 and 23, will leave 5 as the remainder in each case.

SSC MTS 26/07/2022 (Evening)

- (a) 24 (b) 25 (c) 26 (d) 20

SSC CHSL 2021 Tier - 1

Q.98. Four runners started running simultaneously from a point on a circular track. They took 400 seconds, 600 seconds, 720 seconds and 900 seconds to complete one round. After how much time did they meet at the starting point for the first time since the race started?

SSC CHSL 03/06/2022 (Morning)

- (a) 4200 sec (b) 2400 sec
(c) 3600 sec (d) 1800 sec

SSC CGL 2021 Tier - 1

Q.99. Which is the smallest multiple of 7, which leaves 5 as a remainder in each case, when divided by 8, 9, 12 and 15?

SSC CGL 12/04/2022 (Morning)

- (a) 365 (b) 1085 (c) 2525 (d) 725

SSC CGL 2020 Tier - 2

Q.100. Three numbers are in the ratio $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$. If the difference between the greatest number and the smallest number is 33, then HCF of the three numbers is:

SSC CGL Tier II (03/02/2022)

- (a) 9 (b) 5 (c) 13 (d) 11

SSC MTS 2020 Tier - 1

Q.101. The least number of square tiles required to pave the ceiling of a room 12 m 95 cm long and 3 m 85 cm broad, is:

SSC MTS 11/10/2021 (Afternoon)

- (a) 417 (b) 407 (c) 467 (d) 437

SSC CPO 2020 Tier - 1

Q.102. What is the sum of the digits of the least number which when divided by 15, 18 and 36 leaves the same remainder 9 in each case and is divisible by 11 ?
SSC CPO 25/11/2020 (Morning)
(a) 18 (b) 16 (c) 17 (d) 15

SSC CGL 2019 Tier - 2

Q.103. When 1062, 1134 and 1182 are divided by the greatest number x , the remainder in each case is y . What is the value of $(x - y)$?
SSC CGL Tier II (15/11/2020)
(a) 19 (b) 17 (c) 16 (d) 18

Q.104. Let x be the greatest number which when divided by 955, 1027, 1075, the remainder in each case is the same. Which of the following is NOT a factor of x ?
SSC CGL Tier II (16/11/2020)
(a) 6 (b) 16 (c) 4 (d) 8

Answer Key :-

1.(b)	2.(d)	3.(a)	4.(a)
5.(b)	6.(b)	7.(c)	8.(d)
9.(d)	10.(d)	11.(d)	12.(a)
13.(d)	14.(d)	15.(d)	16.(c)
17.(a)	18.(a)	19.(b)	20.(a)
21.(d)	22.(c)	23.(b)	24.(c)
25.(d)	26.(b)	27.(a)	28.(b)
29.(a)	30.(c)	31.(c)	32.(b)
33.(b)	34.(d)	35.(d)	36.(b)
37.(c)	38.(c)	39.(c)	40.(a)
41.(d)	42.(b)	43.(c)	44.(c)
45.(b)	46.(c)	47.(c)	48.(d)
49.(d)	50.(c)	51.(d)	52.(b)
53.(b)	54.(d)	55.(d)	56.(c)
57.(a)	58.(b)	59.(d)	60.(a)
61.(b)	62.(b)	63.(c)	64.(a)
65.(d)	66.(b)	67.(b)	68.(a)
69.(d)	70.(d)	71.(d)	72.(a)
73.(a)	74.(a)	75.(b)	76.(a)
77.(c)	78.(a)	79.(c)	80.(b)
81.(c)	82.(c)	83.(c)	84.(c)
85.(b)	86.(b)	87.(d)	88.(a)
89.(c)	90.(b)	91.(a)	92.(a)
93.(c)	94.(a)	95.(a)	96.(d)
97.(a)	98.(c)	99.(b)	100.(d)
101.(b)	102.(a)	103.(d)	104.(b)

Solutions :-

Sol.1.(b) According to the question,
 $x^2 - 8x + 15$
 $x^2 - (5 + 3)x + 15$
 $x(x - 5) - 3(x - 5)$
 $(x - 5), (x - 3)$
 And $x^2 - 5x + 6$
 $x^2 - (3 + 2)x + 6$
 $x(x - 3) - 2(x - 2)$
 $(x - 3), (x - 2)$
 So the LCM will be $(x - 5), (x - 2), (x - 3)$

Sol.2.(d) According to the question,
 $1 \text{ m} = 100 \text{ cm}$
 HCF of 315, 500, 685 = 5
 $315 = 5 \times 3 \times 21$
 $500 = 5 \times 5 \times 20$
 $685 = 5 \times 137$
 So the greatest possible length be 5 cm

Sol.3.(a)
 $(x^6 + 1) = (x^2 + 1)(x^4 + 1 - x^2)$
 $(x^4 - 1) = (x^2 + 1)(x^2 - 1)$
 H.C.F = $(1 + x^2)$

Sol.4.(a) L.C.M = 533
 As we know that L.C.M of two prime numbers is always the product of the two prime numbers
 So, numbers = 13 and 41
 Hence, $4y - x = 13 \times 4 - 41 = 52 - 41 = 11$

Sol.5.(b)
 $(x^3 - 8) = (x - 2)(x^2 + 2x + 4)$
 $(x^3 - 6x^2 + 12x - 8) = (x - 2)^3$
 and $(x^3 - 4x^2 + 4x) = x(x - 2)^2$
 HCF of $(x^3 - 8), (x^3 - 6x^2 + 12x - 8)$ and $(x^3 - 4x^2 + 4x) = (x - 2)$

Sol.6.(b) Required time = LCM of 50, 75 and 100 $\Rightarrow 300 = 5$ minutes

Sol.7.(c)
 HCF of 7000, 3850 and 12950
 $7000 = 350 \times 20$
 $3850 = 350 \times 11$
 $12950 = 350 \times 37$
 So the maximum length will be = 350 mm or 35 cm

Sol.8.(d) Maximum capacity of container = HCF (850, 680) = 170 litres

Sol.9.(d)
 $(219 - 3), (365 - 5), (511 - 7)$ will be exactly divisible by that two digit number
 Required number = HCF of 216, 360, 504 = 72

Sol.10.(d)
 L.C.M of 5, 10, 15, 20 and 25 = 300 sec.
 Required number = $\frac{80 \times 60}{300} + 1$

= 17 times

Sol.11.(d) According to question,
 Required time when they meet = LCM (15, 18) = 90 min

Sol.12.(a) $x = 224, y = 322$ (given)
 Factors of $23 \times x = 23 \times 224 = 23 \times 2^5 \times 7$
 Factors of $a \times y = a \times 322 = a \times 2 \times 7 \times 23$
 For highest common factor a must be 2^4 because all other factors are already available. $\Rightarrow$ so, possible value of $a = 16$

Sol.13.(d) HCF of 108, 72, $5a$ = Difference of any two no's or the factor of it which divides the given no completely i.e. $108 - 72 = 36 = a$
 Then,
 LCM of 108, 72 and $36 = 36 \times 2 \times 3 = 216$

Sol.14.(d) $554 - 43 = 511, 714 - 57 = 657, 213 - 67 = 146$
 The greatest positive integer = HCF of 511, 657 and 146 = Difference of any two no's or the factors of it which divided the given no's completely i.e.
 $657 - 511 = 146 = 2 \times 73$
 So, the required no = 73

Sol.15.(d) $(4^{315} - 1)(4^{25} - 1)$
 HCF of 315 and 25 = 5
 HCF of $(4^{315} - 1)(4^{25} - 1) = 4^5 - 1 = 1023$

Sol.16.(c) HCF of 120, 240, 150 = 30
 30 fruits can be packed in a carton of each fruits,
 Then the minimum number of cartons is,

$$\text{Carton of apples} = \frac{120}{30} = 4$$

$$\text{Carton of oranges} = \frac{240}{30} = 8$$

$$\text{carton of pears} = \frac{150}{30} = 5$$

Then the minimum number of cartons = $4 + 8 + 5 = 17$

Sol.17.(a) The time when the traffic lights changed simultaneously after every interval = LCM of 48, 72, 108 = 432 sec = 7 min 12 sec
 So, the required time = $8 : 20 + 7 \text{ min } 12 \text{ sec} = 8 : 27 : 12 \text{ a.m}$

Sol.18.(a) since, the given no's are prime no's. So, the HCF of (A, B) = 1 and LCM of (A, B) = $A \times B = AB$
 According to the question,
 $A \times B = 209 = 19 \times 11$
 So, $A = 19$ and $B = 11$ ($\because A > B$)
 Hence, $A^2 - B = 19^2 - 11 = 361 - 11 = 350$

Sol.19.(b) As, HCF of (13, a), (13, b) and (13, c) is 13 where a, b and c must be multiples of 13.
 Only possible value for a, b and c such that $a < b < c < 60$ is 26, 39 and 52.

So, $a = 26$, $b = 39$ and $c = 52$.

$$\frac{a+c}{b} = \frac{26+52}{39} = 2.$$

Sol.20.(a) Let numbers be $81x$ and $81y$.

$$81[x+y] = 1215 \Rightarrow x+y = 15$$

Since numbers lie between 500 and 700 then, the values of x and y are 8 and 7.

So, the no's are 81×8 and 81×7

$$\begin{aligned} \text{Now, } \frac{1}{81 \times 8} + \frac{1}{81 \times 7} \\ = \frac{1}{81} \left[\frac{1}{8} + \frac{1}{7} \right] = \frac{1}{81} \left[\frac{7+8}{56} \right] \\ = \frac{1}{81} \left[\frac{15}{56} \right] = \frac{5}{1512} \end{aligned}$$

Sol.21.(d) In such questions,

Subtract the numbers in the following manner :

$$7306 - 6454 = 852, 8797 - 7306 = 1491$$

$$\text{Factors of } 852 = 4 \times 213$$

$$\text{Factors of } 1491 = 213 \times 7$$

HCF of (852 and 1491) = 213, Which will be the divisor for any of the given no i.e. 6454, 7306 and 8797

So, Remainder = 64,

$$\therefore d - r = 213 - 64 = 149$$

Sol.22.(c)

$$\begin{array}{r} 1785 \overline{) 1995} (1 \\ \underline{-1785} \\ 210 \overline{) 1785} (8 \\ \underline{-1680} \\ 105 \overline{) 210} (2 \\ \underline{-210} \\ \text{xxx} \end{array}$$

In such cases the last remainder is always zero.

The required two numbers are 1995 and 1785

Sum of these two numbers

$$= 1995 + 1785 = 3780$$

Sol.23.(b) LCM of (5,6,8,9,12) = 360

So, the number can be written as $360k + 1$, divisible by 13.

So, putting $k = 10$, we have ;

$$x = 360 \times 10 + 1 = 3601$$

Now, When 3601 is divided by 31, the remainder will be 5.

Sol.24.(c) Formula used :

$$A^3 + B^3 = (A+B)\{(A+B)^2 - 3AB\}$$

$$A+B = p+q \text{ (given)}$$

We know that ,

HCF $\times$ LCM = product of the two no's

$$\Rightarrow A \times B = p \times q$$

$$\text{Now, } A^3 + B^3 = (p+q)\{(p+q)^2 - 3pq\}$$

$$\Rightarrow p^3 + q^3$$

Sol.25.(d) $460 - 26 = 434$, $491 - 26 = 465$

$$553 - 26 = 527$$

Now, HCF of 434, 465 and 527

$$434 = 2 \times 7 \times 31 \Rightarrow 465 = 3 \times 5 \times 31$$

$$527 = 17 \times 31, \text{ So, the required HCF} = 31$$

Sol.26.(b) $3 - 1 = 2$, $4 - 2 = 2$ and $5 - 3 = 2$
smallest 3 digit number = 100

LCM of 3,4 and 5 = 60

$$\begin{array}{r} 60 \overline{) 100} (1 \\ \underline{-60} \\ 40 \end{array}$$

Smallest three digit number

$$= 100 + (60 - 40) - 2 = 118$$

$$\text{Sum of the digits} = 1 + 1 + 8 = 10$$

Sol.27.(a) Let the numbers be $13x$ and $13y$. According to question

$$13x \times 13y = 6760 \Rightarrow 169xy = 6760$$

$$\Rightarrow xy = 40$$

Required number of pairs are (1, 40) and (5, 8)

$$\text{Sol.28.(b)} \quad 432 = 2^4 \times 3^3,$$

$$594 = 2 \times 3^3 \times 11, 702 = 2 \times 3^3 \times 13$$

$$\text{HCF} = 2 \times 3^3 = 54$$

$$\text{Cans for first variety of oil} = \frac{432}{54} = 8$$

$$\text{Cans for 2nd variety of oil} = \frac{594}{54} = 11$$

$$\text{Cans for 3rd variety of oil} = \frac{702}{54} = 13$$

Sol.29.(a) From the given statements only option (a) is true. As LCM of two numbers is divisible by their HCF.

Sol.30.(c)

For same quantity of milk = HCF of (75, 45, 30) = 15 liter

$$\text{Trays for cow milk} = \frac{75}{15} = 5$$

$$\text{Trays for toned milk} = \frac{45}{15} = 3$$

$$\text{Trays for full cream milk} = \frac{30}{15} = 2$$

$$\text{Total trays} = 5 + 3 + 2 = 10$$

The least number of trays required to place all the milk is 10.

Sol.31.(c)

Let three numbers are $5x$, $7x$, $9x$

According to question ,

$$(5 \times 7 \times 9) x = 1260$$

$$\Rightarrow 315x = 1260 \Rightarrow x = 4$$

$$\therefore \text{Three numbers are } 20, 38, 36$$

$$\text{Required difference} = 36 - 20 = 16$$

Sol.32.(b) LCM(4, 6, 8) = 24

It means that in every 24 minute red light blinks

Now, 3 hours = 180 minute

So, in the interval of 3 hours red light

$$\text{blinks} = \frac{180}{24} = 7 \text{ times}$$

(consider integer value only)

Sol.33.(b)

Amita, Amrita and Tosita went office on 2nd, 3rd and 6th day respectively

$$\text{LCM}(2, 3, 6) = 6$$

So, on every 6th day all three attend the office

Now, total days in last two months = $30 + 31 = 61$ days

Hence, no. of times they attend together the office in last two months = $\frac{61}{6} = 10$ times (only integer value considered)

Sol.34.(d) LCM (20, 25, 35, 40)

$$= 2^3 \times 5 \times 5 \times 7 = 1400$$

Now, in every case remainder is 6 less than the divisor

$$\text{So, required no.} = 1400 - 6 = 1394$$

Sol.35.(d) Given, $x = a^2 b^3$ and $y = a^3 b$

$$\text{HCF}(x, y) = a^2 b$$

Sol.36.(b) According to question,

$$\text{Given: } x = 2k, y = 3m, z = 4n.$$

$$\text{Then, } 6x = 12k, 4y = 12m, 3z = 12n.$$

All are multiples of 12.

$$\text{HCF}(6x, 4y, 3z) = 12.$$

Hence, the HCF(6x, 4y, 3z) is at least 12.

Sol.37.(c) Given, $3^3 \times 11^2, 2^4 \times 3^2 \times 5^3 \times 11^3$ and $3 \times 5 \times 7 \times 11$

$$\text{HCF} = 3 \times 11 = 33$$

Sol.38.(c)

Let the number be $7x$ and $9x$ respectively,

According to question,

$$\Rightarrow 7 \times 9 \times x = 630 \Rightarrow x = 10$$

So, the number will be 70 and 90.

Sol.39.(c) HCF of (156 and 96) = 12

According to question,

$$\Rightarrow 156 \times 12 + 96A = 12$$

$$\Rightarrow 1872 + 96A = 12$$

$$\Rightarrow A = \frac{12 - 1872}{96} = \frac{-1860}{96} = \frac{-155}{8}$$

$$= -19.375$$

Sol.40.(a) Numbers = 1.75, 5.6, 7

$$= \frac{175}{100}, \frac{560}{100}, \frac{700}{100}$$

$$\text{HCF} = \frac{\text{HCF of } (175, 560, 700)}{\text{LCM of } (100, 100, 100)} = \frac{35}{100} = 0.35$$

Sol.41.(d)

$$\text{Factor of } (x^3 - 27) = (x-3)(x^2 + 9 + 3x)$$

$$\text{Factor of } (x^2 - x - 6) = (x+2)(x-3)$$

$$\text{Factor of } (x+2) = (x+2)$$

$$\text{LCM of } (x^3 - 27), (x^2 - x - 6) \text{ and } (x+2) = (x-3)(x+2)(x^2 + 9 + 3x)$$

Sol.42.(b) Product of two numbers

$$= \text{LCM} \times \text{HCF} = 24$$

From the options we can see that only option (b) has numbers which can give 24 as product.

Hence, two numbers are 6 and 4.

Sol.43.(c) L.C.M. of 1, 4, 5, 7, 8 = 280

$$\text{H.C.F. of } 6, 15, 9, 27, 3 = 3$$

L.C.M. of fractions

$$= \frac{\text{L.C.M. of numerator}}{\text{H.C.F. of denominator}} = \frac{280}{3} = 93 \frac{1}{3}$$

Sol.44.(c) L.C.M of 17 and 36 = 612
Largest number of 4-digit = 9999
When we divide 9999 by 612, we get 207 as a remainder.
Required number = 9999 - 207 = 9792

Sol.45.(b) Let the numbers are $3x$ and $4x$
According to question,
 $\Rightarrow 3x \times 4x = 2700 \Rightarrow x = 15$
So, the sum of the numbers
 $= 15 \times 3 + 15 \times 4 = 45 + 60 = 105$
Short Trick :- $(3x + 4x) = 7x$
Hence, the sum of the numbers must be a multiple of 7.
Now, from options, 105 is the required answer.

Sol.46.(c) According to the question,
 $261 - 5 = 256 \Rightarrow 16 \times 16$
 $853 - 5 = 848 \Rightarrow 16 \times 53$
 $1221 - 5 = 1216 \Rightarrow 16 \times 76$
So the HCF of 256, 848, 1216 will be 16
So the required number will be 16

Sol.47.(c) According to the question,
LCM of $\frac{3}{2}, \frac{81}{16}, \frac{9}{8} = \frac{LCM}{HCF} = \frac{81}{2}$

Sol.48.(d) According to the question,
1 m = 100 cm
HCF of (600, 525, 1250) = 25 cm or 0.25 m
So the greatest possible length will be 0.25 m

Sol.49.(d) L.C.M of 2, 4, 6 and 8 = 24
When we divide 1351 by 24 we get 7 as remainder.
So, the required number = 24 - 7 = 17

Sol.50.(c) H.C.F of m and $n = a$
Given, $n = ab$
L.C.M = $\frac{\text{product of numbers}}{H.C.F}$
 $= \frac{m \times ab}{a} = mb$

Sol.51.(d) According to the question,
 $1105 = 17 \times 1^{\text{st}} \text{ number}$
 $\Rightarrow 1^{\text{st}} \text{ number} = 65$

Formula:- 1st no. $\times$ 2nd no. = LCM $\times$ HCF
 $65 \times 2^{\text{nd}} \text{ no.} = 1105 \times 5$
2nd no. = 85
So the no. will be 65 and 85

Sol.52.(b) HCF(45, 55) = 5
According to the question,
 $5 = 55 \times 5 + 45m$ ----- (1)
By go through the options one by one
Value $m = -6$ satisfied the equations
i.e. $5 = 55 \times 5 + 45 \times (-6)$
 $\Rightarrow 5 = 275 - 270 = 5$

Sol.53.(b) LCM(4, 9, 12, 15) = 180
hence, least number = 180 + 3 = 183

Sol.54.(d)
Let numbers be $13x$ and $15x$ respectively,

Then, HCF($13x, 15x$) = x
According to the question,
 $x = 12$
So, Numbers = 156 and 180
Hence, LCM(156, 180) = 2340

Sol.55.(d) Remainder = HCF(6, 8, 9) = 1
And LCM(6, 8, 9) = 72
So, number should be = $72k + 1$
After putting $k = 6$ we get;
Number = $72 \times 6 + 1 = 433$

Sol.56.(c)
L.C.M of 20, 28, 34, 60 and 75 = 35700
So, the required least number = 35700

Sol.57.(a)
H.C.F of 1036, 1813 and 3885 = 259
Hence, the required largest number = 259

Sol.58.(b) According to the question,
LCM of $\frac{15}{100}, \frac{18}{100}, \frac{45}{100} = \frac{90}{100} = 0.9$

Sol.59.(d) According to the question,
LCM of 12, 14, 16 and 18 be 1008
So the 504 is the least number when it doubles will be exactly divisible by the 12, 14, 16, and 18.

Sol.60.(a) Given,
LCM = $5 \times$ HCF
Let numbers be "a" and "b"
 $a \times b = 20480$
HCF $\times$ LCM = $a \times b$
HCF $\times 5 \times$ HCF = 20480
 $HCF^2 = \frac{20480}{5} = 4096 \Rightarrow HCF = 64$
LCM = $5 \times$ HCF = $5 \times 64 = 320$

Sol.61.(b) Let numbers be a and b
 $a \times b = \text{HCF} \times \text{LCM}$
 $a \times b = 5 \times 60$ ---- (1)
 $a + b = 60$ ---- (2)
Now,
 $\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{60}{5 \times 60} = \frac{1}{5}$

Sol.62.(b) According to the question,
LCM of 9, 12, 15, 25 and 27 will be 2700
So the smallest 5 digit number which exactly divisible by these numbers
 $= 2700 \times 4 = 10800$

Sol.63.(c) According to the question,
1st number = $627 - 2 = 625$
2nd number = $15630 - 5 = 15625$
3rd number = $3128 - 3 = 3125$
HCF of 625, 15625 and 3125 = 625.
So the required number will be 625.

Sol.64.(a) According to the question,
No. will be = $4 \times 57, 5 \times 57, 6 \times 57$
 $= 228, 285, 342$

Sol.65.(d) Let three numbers be $3x, 5x$ and $7x$ respectively,

Then, LCM ($3x, 5x, 7x$) = $105x$
 $105x = 2625 \Rightarrow x = 25$
numbers = $(3 \times 25), (5 \times 25), (7 \times 25)$
So, HCF = 25

Sol.66.(b) LCM (3, 5, 9) = 45
Now, remainder $\frac{999}{45} = 9$
So, required largest three digit number
 $= (999 - 9) + 2 = 992$

Sol.67.(b) According to the question,
If the HCF of any 2 numbers are x then the LCM of their will be the multiple of x .
So in option (b) 74 is not the multiple of 12 so option (b) is the right answer

Sol.68.(a)
Let the numbers be $37x$ and $37y$
According to the question,
 $(37 \times x) \times (37 \times y) = 4107$
 $xy = 3$ {where x and y be co-prime}
So, possible value of x and y be 1 and 3
Hence, the greater number = $37 \times 3 = 111$

Sol.69.(d)
LCM $\times$ HCF = 1st number $\times$ 2nd number
 $1440 \times 32 = 288 \times 2^{\text{nd}} \text{ number}$
2nd number = 160

Sol.70.(d) LCM of 4, 8, 9 = 72
Greatest three digit number = 999
 $\frac{999}{72} = \text{Rem (63)}$
Required number = $999 - 63 = 936$

Sol.71.(d) The smallest number which is completely divisible by 12, 14 and 18 will be the LCM of these numbers.
So the LCM of 12, 14 and 18 = 252
Required number = $252 - 5 = 247$

Sol.72.(a) Let the numbers are $3x$ and $4x$
According to the question,
 $3x + 4x = 70 \Rightarrow x = 10$
Numbers 30 and 40
LCM of 30 and 40 = 120

Sol.73.(a) L.C.M of 2, 3, 5, 6, 9 and 18 = 90
 $90 = 2 \times 3 \times 3 \times 5$
As 2 and 5 don't have any pair
So,
the required number = $90 \times 2 \times 5 = 900$

Sol.74.(a)
L.C.M. (12, 16, 24, 28) = $3 \times 4 \times 4 \times 7$
 $= 336$
Smallest 6 digit no. = 100000
Rem. $(\frac{100000}{336}) = 208$

So, required smallest 6 digit no.
 $= 100000 + (336 - 208) = 100128$

Sol.75.(b) Let,
L.C.M. be 20 units then H.C.F. be 1 unit.
According to the question,
Difference (19 units) = $456 \Rightarrow 1 \text{ unit} = 24$

Now, L.C.M. $\times$ H.C.F. = 1st no. $\times$ 2nd no.
 $(20 \times 24) \times (1 \times 24) = 96 \times 24$

$$\Rightarrow 2\text{nd no.} = \frac{(20 \times 24) \times (1 \times 24)}{96} = 120$$

Sol.76.(a) The HCF of two prime numbers is always 1.

Sol.77.(c) LCM (5, 6, 12) = 60

$$\text{Remainder } \frac{478}{60} = 58$$

Hence,

$$\text{Required least number} = 60 - 58 = 2$$

Sol.78.(a)

Product of two co - prime number = 483

HCF of co - prime numbers always be 1

So, least common multiple of both numbers = $1 \times 483 = 483$

Sol.79.(c) $14 = 2 \times 7$, $84 = 2^2 \times 3 \times 7$

$$28 = 2^2 \times 7$$

Required perfect square no.

$$= 2^2 \times 3^2 \times 7^2 = 1764$$

Sol.80.(b) According to the question,

Number that completely divisible,

$$456 - 6 = 450 \Rightarrow 553 - 3 = 550$$

According to the option 50 is the largest number that divides 450 and 550 completely.

Sol.81.(c) The number which is divided

$$\text{by } 14, 18 \text{ and } 36 \Rightarrow \text{LCM}(14, 18, 36) = 252$$

Remainder = 1{given}

So, Required number = 253

Sol.82.(c) Least common multiple of

$$12, 24 \text{ and } 36 = 72$$

Highest common multiple of

$$12, 24 \text{ and } 36 = 12$$

$$\text{Required ratio} = 72 : 12 = 6 : 1$$

Sol.83.(c) According to question,

$$\text{LCM}(10, 15, 20) = 60$$

From the given options,

$$\text{option (c) } (60 \times 15) = 900,$$

is the correct answer.

Sol.84.(c) $513 = 3 \times 3 \times 57$

$$1017 = 3 \times 3 \times 113$$

$$\text{So, H.C.F.}(513, 1017) = 3 \times 3 = 9$$

Sol.85.(b) Factor of $40 = 2 \times 2 \times 2 \times 5$

By the option 25 never be the HCF

Sol.86.(b) Smallest number that is

divisible by 8 and 12 will be 24

$$\text{Required number} = 24 \times 6 = 144$$

Sol.87.(d) Let the numbers be $10x$ and $10y$

$$\text{Then, } 10x \times 10y = 1500 \Rightarrow xy = 15$$

Possible value of x and y are

$$\Rightarrow (1, 15) \text{ and } (3, 5)$$

Sol.88.(a) Let, the no. are $15a$ and $15b$.

$$36x^2 - 81y^2 = 9(2x - 3y)(2x + 3y)$$

$$= 9(2 \times 15a - 3 \times 15b)(2 \times 15a + 3 \times 15b)$$

$$= 9 \times 15\{(2a - 3b)(2a + 3b)\}$$

$$\text{And, } 81x^2 - 9y^2 = 9(3x - y)(3x + y)$$

$$= 9(3 \times 15a - 15b)(3 \times 15a + 15b)$$

$$= 9 \times 15\{(3a - b)(3a + b)\}$$

$$\text{Their H.C.F.} = 135$$

$$\text{Sol.89.(c) H.C.F. } \left(\frac{12}{5}, \frac{14}{15}, \frac{16}{17}\right)$$

$$= \frac{\text{H.C.F.}(12, 14, 16)}{\text{L.C.M.}(5, 15, 17)} = \frac{2}{255}$$

Sol.90.(b) H.C.F of two or more numbers is the highest number which exactly divides all the given numbers.

Example:- H.C.F of 4 and 6 = 2

2 exactly divides 4 and 6.

Sol.91.(a) LCM of a and $b = 42$

Then, LCM of $5a$ and $11b = 55ab$

$$= 55 \times 42 = 2310$$

Sol.92.(a) Let the two no's be $68x$ and $68y$ where x and y are co-prime no's

ATQ,

$$68x + 68y = 1224 \Rightarrow 68(x + y) = 1224$$

$$x + y = \frac{1224}{68} = 18$$

Required no of pairs satisfying the above conditions = 3 i.e. (1, 17); (5, 13); (7, 11)

Sol.93.(c)

Let the numbers are $3x$, $7x$ and $11x$

$$\text{L.C.M} = 1386$$

$$\Rightarrow (3 \times 7 \times 11)x = 1386$$

$$\Rightarrow 231x = 1386 \Rightarrow x = 6$$

So, the numbers are 18, 42 and 66.

Hence, the sum of the largest and the least number = $66 + 18 = 84$

Sol.94.(a)

$$P = 2^8 \times 3^5, Q = 2^3 \times 3^4, R = 3^5 \times 2^7$$

$$\text{HCF of } P, Q \text{ and } R = 2^3 \times 3^4$$

Sol.95.(a) Let two numbers are $4a$ and

$4b$ as their HCF is given as 4

LCM of $4a$ and $4b$ is $4ab$

Now according to the question

$$\frac{\text{LCM}}{\text{Sum of numbers}} = \frac{4ab}{4(a + b)} = \frac{12}{7}$$

$$ab = 12 \text{ and } a + b = 7,$$

by hit and trial $\Rightarrow a = 4$ and $b = 3$

the numbers are $4a$ and $4b = 16$ and 12

$$\text{Products of two numbers} = 16 \times 12 = 192$$

Sol.96.(d) A number that is divisible only by itself and 1 is called prime number

Highest common factor (HCF) of K_1 and

$$K_2 = 1$$

And least common multiple (LCM) -

$$= K_1 \times K_2$$

$$\text{Then the product} = K_1 \times K_2$$

Sol.97.(a) LCM of 7, 11 and 23 = 1771

Remainder in each case = 5

$$\text{So, required number} = 1771 + 5 = 1776$$

$$\text{Value of } x = 1800 - 1776 = 24$$

Sol.98.(c) First time of meeting of all of them running on a circular track = LCM of (400, 600, 720, 900)

$$400 = 2^4 \times 5^2, 600 = 2^3 \times 3 \times 5^2$$

$$720 = 2^4 \times 3^2 \times 5, 900 = 3^2 \times 5^2 \times 2^2$$

$$\text{LCM of } (400, 600, 720, 900)$$

$$= 2^4 \times 3^2 \times 5^2 = 3600 \text{ sec}$$

Sol.99.(b) LCM (8, 9, 12, 15) = 360

So, the number is $360k + 5$, which is divisible by 7 $\Rightarrow$ Putting $k = 3$, we have ;

$$\text{Required no} = 360 \times 3 + 5$$

$$= 1080 + 5 = 1085$$

Sol.100.(d) According to the question,

$$\frac{1}{2} : \frac{2}{3} : \frac{3}{4} = 6 : 8 : 9$$

Let three numbers be $6x$, $8x$ and $9x$

$$\text{So, } 9x - 6x = 33 \Rightarrow 3x = 33 \Rightarrow x = 11$$

$$\text{HCF } (6x, 8x, 9x) = x = 11$$

Sol.101.(b) HCF of (1295, 385) = 35

$$\text{Area of each square tile} = (35 \times 35) \text{ cm}^2$$

So, least number of square tiles required

$$= \frac{1295 \times 385}{35 \times 35} = 407$$

Sol.102.(a) LCM of (15, 18, 36) = 180

Number can be written as $180k + 9$, which is divisible by 11.

Now, putting $K = 6$ we get :

$$180 \times 6 + 9 = 1080 + 9 = 1089$$

$$\therefore \text{Required Sum} = 1 + 0 + 8 + 9 = 18$$

Sol.103.(d) To find the required greatest number, we need to find the HCF of the difference between all the three given numbers.

$$1134 - 1062 = 72, 1182 - 1134 = 48$$

$$\text{HCF of } 72 \text{ and } 48 = 24$$

Now, When 1062 is divided by 24 leaves, remainder 6.

$$\text{So, } x - y = 24 - 6 = 18$$

Sol.104.(b) To find the value of x , we need to find the HCF of the difference between all the three given numbers.

$$1027 - 955 = 72, 1075 - 1027 = 48$$

$$\text{HCF of } (72 \text{ and } 48) = 24$$

$$\text{Factors of } 24 = 1, 2, 3, 4, 6, 8, 12, 24$$

Hence, 16 is not the factor of x